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Cell Cycle and Cell Division

NEET Biology · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer)	105 questions
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MCQ (Single Correct Answer) 105 questions

Q1. RE-NEET 2026

Open solution

Given below are two statements :

Statement I : Chromosomes are fully condensed at the end of prophase I.

Statement II : Meiosis I resembles mitosis.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is incorrect, but Statement II is true

(B) Both Statement I and Statement II are true

(C) Both Statement I and Statement II are false

(D) Statement I is correct, but Statement II is false

Q2. RE-NEET 2026

Open solution

The correct sequence of adult cell cycle phases is ____ .

(A) S-M-G2-G1 (B) G1-G2-S-M (C) G1-M-G2-S (D) G1-S-G2-M

Q3. NEET 2026

Open solution

Match List I with List II :

List-I (Phase of cell cycle)		List-II (Activity)	
A.	G_1 phase	I.	Actual cell division occurs
B.	S phase	II.	Cell is metabolically active and continuously grows but does not replicate its DNA
C.	G_2 Phase	III.	Synthesis of DNA occurs and the amount of DNA per cell doubles
D.	M phase	IV.	Proteins are synthesized while cell growth continues

Choose the correct answer from the options given below :

(A) A-IV, B-I, C-II, D-III (B) A-I, B-II, C-III, D-IV

(C) A-III, B-IV, C-I, D-II (D) A-II, B-III, C-IV, D-I

Q4. NEET 2025

Open solution

What is the main function of the spindle fibers during mitosis?

(A) To repair damaged DNA (B) To regulate cell growth

(C) To separate the chromosomes (D) To synthesize new DNA

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Q5. NEET 2024 (Re-Examination)

Open solution

Given below are two statements:

Statement I: Failure of segregation of chromatids during cell cycle resulting in the gain or loss of whole set of chromosome in an organism is known as aneuploidy.

Statement II: Failure of cytokinesis after anaphase stage of cell division results in the gain or loss of a chromosome is called polyploidy.

In the light of the above statements, choose the correct answer from the options given below:

(A) Statement I is true but Statement II is false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are true

(D) Both Statement I and Statement II are false

Q6. NEET 2024 (Re-Examination)

Open solution

Recombination between homologous chromosomes is completed by the end of

(A) Diakinesis **(B)** Zygotene **(C)** Diplotene **(D)** Pachytene

Q7. NEET 2024 (Re-Examination)

Open solution

Match List-I with List-II :

	List - I		List - II
(A)	Metacentric chromosome	(I)	Chromosome has a terminal centromere
(B)	Sub-metacentric chromosome	(II)	Middle centromere forming two equal arms of chromosome
(C)	Acrocentric chromosome	(III)	Centromere is slightly away from the middle of chromosome resulting into two unequal arms
(D)	Telocentric chromosome	(IV)	Centromere is situated close to its end forming one extremely short and one very long arm

Choose the correct answer from the options given below:

(A) A-II, B-I, C-IV, D-III **(B)** A-IV, B-I, C-II, D-III

(C) A-I, B-II, C-III, D-IV **(D)** A-II, B-III, C-IV, D-I

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Q8. NEET 2024 (Re-Examination)

Open solution

Match List-I with List-II.

	List - I Event		List - II Stage of Prophase-I (Meiosis-I)
(A)	Chiasmata formation	(I)	Pachytene
(B)	Crossing over	(II)	Diakinesis
(C)	Synaptonemal complex formation	(III)	Diplotene
(D)	Terminalisation of chiasmata	(IV)	Zygotene

Choose the correct answer from the options given below :

(A) A-III, B-I, C-IV, D-II
(B) A-II, B-I, C-III, D-IV
(C) A-III, B-I, C-II, D-IV
(D) A-II, B-III, C-IV, D-I

Q9. NEET 2024 (Re-Examination)

Open solution

Match List-I with List-II.

	List - I		List - II
(A)	Cells are metabolically active and proliferate	(I)	G_2 phase
(B)	DNA replication takes place	(II)	G_1 phase
(C)	Proteins are synthesised	(III)	G_0 phase
(D)	Quiescent stage with metabolically active cells	(IV)	S phase

Choose the correct answer from the options given below:

(A) A-IV, B-II, C-III, D-I
(B) A-I, B-III, C-IV, D-II
(C) A-I, B-I, C-III, D-IV
(D) A-II, B-IV, C-I, D-III

Q10. NEET 2024

Open solution

Spindle fibers attach to kinetochores of chromosomes during

(A) Prophase **(B)** Metaphase **(C)** Anaphase **(D)** Telophase

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Q11. NEET 2024

Open solution

Given below are two statements:

Statement I : Chromosomes become gradually visible under light microscope during leptotene stage.

Statement II : The beginning of diplotene stage is recognized by dissolution of synaptonemal complex.

In the light of the above statements, choose the correct answer from the options given below:

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q12. NEET 2024

Open solution

Following are the stages of cell division :

A. Gap 2 phase

B. Cytokinesis

C. Synthesis phase

D. Karyokinesis

E. Gap 1 phase

Choose the correct sequence of stages from the options given below :

(A) C-E-D-A-B **(B)** E-B-D-A-C **(C)** B-D-E-A-C **(D)** E-C-A-D-B

Q13. NEET 2024

Open solution

Match List I with List II :

	List - I (Sub Phases of Prophase I)		**List - II (Specific Characters)**
A.	Diakinesis	I.	Synaptonemal complex formation
B.	Pachytene	II.	Completion of terminalisation of chiasmata
C.	Zygotene	III.	Chromosomes look like thin threads
D.	Leptotene	IV.	Appearance of recombination nodules

Choose the correct answer from the options given below

(A) A-IV, B-II, C-III, D-I **(B)** A-I, B-II, C-IV, D-III

(C) A-II, B-IV, C-I, D-III **(D)** A-IV, B-III, C-II, D-I

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Q14. NEET 2023 Manipur

The dissolution of synaptonemal complex occurs during :

(A) Pachytene (B) Diplotene (C) Diakinesis (D) Leptotene

Open solution

Q15. NEET 2023 Manipur

Doubling of the number of chromosomes can be achieved by disrupting mitotic cell division soon after :

(A) Anaphase (B) Telophase (C) Prophase (D) Metaphase

Open solution

Q16. NEET 2023 Manipur

During which stages of mitosis and meiosis, respectively does the centromere of each chromosome split ?

(A) Metaphase, Metaphase II (B) Prophase, Telophase I

(C) Telophase, Anaphase I (D) Anaphase, Anaphase II

Open solution

Q17. NEET 2023 Manipur

Which one of the following is the quiescent stage of cell cycle?

(A) M (B) G_2 (C) G_1 (D) G_0

Open solution

Q18. NEET 2023 Manipur

Select correct sequence of substages of Prophase-I of Meiotic division :

(A) Zygotene

(B) Pachytene

(C) Diakinesis

(D) Leptotene

(E) Diplotene

Choose the correct answer from the options given below :

(A) (D), (B), (A), (E), (C) (B) (A), (B), (D), (E), (C)

(C) (D), (A), (B), (E), (C) (D) (A), (D), (B), (C), (E)

Open solution

Q19. NEET 2023

The process of appearance of recombination nodules occurs at which sub stage of prophase I in meiosis?

(A) Pachytene (B) Diplotene (C) Diakinesis (D) Zygotene

Open solution

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Q20. NEET 2023

Which of the following stages of meiosis involves division of centromere?

(A) Metaphase II (B) Anaphase II (C) Telophase (D) Metaphase I

Open solution

Q21. NEET 2023

Among eukaryotes, replication of DNA takes place in :

(A) S phase (B) G_1 phase (C) G_2 phase (D) M phase

Open solution

Q22. NEET 2023

Match List I with List II:

	List I		List II
(A)	M Phase	(I)	Proteins are synthesized
(B)	G_2 Phase	(II)	Inactive phase
(C)	Quiescent stage	(III)	Interval between mitosis and initiation of DNA replication
(D)	G_1 Phase	(IV)	Equational division

Choose the correct answer from the options given below :

(A) A-IV, B-II, C-I, D-III (B) A-IV, B-I, C-II, D-III

(C) A-II, B-IV, C-I, D-III (D) A-III, B-II, C-IV, D-I

Open solution

Q23. NEET 2023

Select the correct statements.

A. Tetrad formation is seen during Leptotene.

B. During Anaphase, the centromeres split and chromatids separate.

C. Terminalization takes place during Pachytene.

D. Nucleolus, Golgi complex and ER are reformed during Telophase.

E. Crossing over takes place between sister chromatids of homologous chromosome.

Choose the correct answer from the options given below:

(A) B and D only (B) A, C and E only (C) B and E only (D) A and C only

Open solution

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Q24. NEET 2023

Open solution

Given below are two statements:

Statement I : During G_0 phase of cell cycle, the cell is metabolically inactive.

Statement II : The centrosome undergoes duplication during S phase of interphase.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Both Statement I and Statement II are incorrect.

(B) Statement I is correct but Statement II is incorrect.

(C) Statement I is incorrect but Statement II is correct.

(D) Both Statement I and Statement II are correct.

Q25. NEET 2022 Phase 2

Open solution

Which stage of meiosis can last for months or years in the oocytes of some vertebrates?

(A) Diakinesis (B) Leptotene (C) Pachytene (D) Diplotene

Q26. NEET 2022 Phase 2

Open solution

Identify the correct sequence of events during Prophase I of meiosis:

(a) Synapsis of homologous chromosomes

(b) Chromosomes become gradually visible under microscope

(c) Crossing over between non-sister chromatids of homologous chromosomes

(d) Terminalisation of chiasmata

(e) Dissolution of synaptonemal complex

Choose the correct answer from the options given below:

(A) (a), (c), (d), (e), (b) (B) (a), (b), (c), (d), (e)

(C) (b), (c), (d), (e), (a) (D) (b), (a), (c), (e), (d)

Q27. NEET 2022 Phase 2

Open solution

Bivalent or Tetrad formation is a characteristic feature observed during

(A) Chiasmata in zygotene stage (B) Synaptonemal complex in zygotene stage

(C) Chiasmata in Diplotene stage (D) Synaptonemal complex in Pachytene stage

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Q28. NEET 2022 Phase 2

With respect to metaphase, which of the following statements is incorrect?

(A) Chromosomes lie at the equator of the cell

(B) Complete disintegration of nuclear envelope takes place

(C) Chromosomes are highly condensed

(D) Metaphase chromosomes are made up of four sister chromatids held together by centromere

Open solution

Q29. NEET 2022 Phase 1

Which one of the following never occurs during mitotic cell division?

(A) Spindle fibres attach to kinetochores of chromosomes

(B) Movement of centrioles towards opposite poles

(C) Pairing of homologous chromosomes

(D) Coiling and condensation of the chromatids

Open solution

Q30. NEET 2022 Phase 1

The appearance of recombination nodules on homologous chromosomes during meiosis characterizes:

(A) Synaptonemal complex

(B) Bivalent

(C) Sites at which crossing over occurs

(D) Terminalization

Open solution

Q31. NEET 2022 Phase 1

Regarding Meiosis, which of the statements is incorrect?

(A) There are two stages in Meiosis, Meiosis-I and II

(B) DNA replication occurs in S phase of Meiosis-II

(C) Pairing of homologous chromosomes and recombination occurs in Meiosis-I

(D) Four haploid cells are formed at the end of Meiosis-II

Open solution

Q32. NEET 2022 Phase 1

Select the incorrect statement with reference to mitosis :

(A) All the chromosomes lie at the equator at metaphase

(B) Spindle fibres attach to centromere of chromosomes

(C) Chromosomes decondense at telophase

(D) Splitting of centromere occurs at anaphase

Open solution

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Q33. NEET 2021

Which of the following stages of meiosis involves division of centromere?

(A) Telophase II (B) Metaphase I (C) Metaphase II (D) Anaphase II

Open solution

Q34. NEET 2021

Match List-I with List-II.

Open solution

	List-I		List-II
(a)	S phase	(i)	Proteins are synthesized
(b)	G_2 phase	(ii)	Inactive phase
(c)	Quiescent stage	(iii)	Interval between mitosis and initiation of DNA replication
(d)	G_1 phase	(iv)	DNA replication

Choose the correct answer from the options given below.

(A)

(a)	(b)	(c)	(d)
(ii)	(iv)	(iii)	(i)

(B)

(a)	(b)	(c)	(d)
(iii)	(ii)	(i)	(iv)

(C)

(a)	(b)	(c)	(d)
(iv)	(ii)	(iii)	(i)

(D)

(a)	(b)	(c)	(d)
(iv)	(i)	(ii)	(iii)

Q35. NEET 2021

The fruit fly has 8 chromosomes (2n) in each cell. During interphase of Mitosis if the number of chromosomes at G_1 phase is 8, what would be the number of chromosomes after S phase?

(A) 32 (B) 8 (C) 16 (D) 4

Open solution

Q36. NEET 2021

Which stage of meiotic prophase shows terminalisation of chiasmata as its distinctive feature?

(A) Pachytene (B) Leptotene (C) Zygotene (D) Diakinesis

Open solution

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Q37. NEET 2021

The centriole undergoes duplication during :

(A) G_2 phase (B) S-phase (C) Prophase (D) Metaphase

Open solution

Q38. NEET 2020 Phase 2

In a mitotic cycle, the correct sequence of phases is

(A) G_1, G_2, S, M (B) S, G_1, G_2, M

(C) G_1, S, G_2, M (D) M, G_1, G_2, S

Open solution

Q39. NEET 2020 Phase 2

Attachment of spindle fibers to kinetochores of chromosomes becomes evident in :

(A) Metaphase (B) Anaphase (C) Telophase (D) Prophase

Open solution

Q40. NEET 2020 Phase 2

During Meiosis I, in which stage synapsis takes place ?

(A) Leptotene (B) Pachytene (C) Zygotene (D) Diplotene

Open solution

Q41. NEET 2020 Phase 2

Match the following events that occur in their respective phases of cell cycle and select the correct option :

(a) G_1 phase	(i) Cell grows and organelle duplication
(b) S phase	(ii) DNA replication and chromosome duplication
(c) G_2 phase	(iii) Cytoplasmic growth
(d) Metaphase in M - phase	(iv) Alignment of chromosomes

(A) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv) (B) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

(C) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii) (D) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

Open solution

Q42. NEET 2020 Phase 1

Some dividing cells exit the cell cycle and enter vegetative inactive stage. This is called quiescent stage (G_0). This process occurs at the end of

(A) G_1 phase (B) S phase (C) G_2 phase (D) M phase

Open solution

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Q43. NEET 2020 Phase 1

Match the following with respect to meiosis

(A) Zygotene (i) Terminalization
(B) Pachytene (ii) Chiasmata
(C) Diplotene (iii) Crossing over
(D) Diakinesis (iv) Synapsis

Select the correct option from the following

(A) (A) → (iv), (B) → (iii), (C) → (ii), (D) → (i)
(B) (A) → (iii), (B) → (iv), (C) → (i), (D) → (ii)
(C) (A) → (i), (B) → (ii), (C) → (iv), (D) → (iii)
(D) (A) → (ii), (B) → (iv), (C) → (iii), (D) → (i)

Open solution

Q44. NEET 2020 Phase 1

Dissolution of the synaptonemal complex occurs during

(A) Diplotene **(B)** Zygotene **(C)** Pachytene **(D)** Leptotene

Open solution

Q45. NEET 2020 Phase 1

Identify the correct statement with regard to G_1 phase (Gap 1) of interphase

(A) Cell is metabolically active, grows but does not replicate its DNA.
(B) DNA synthesis or replication takes place.
(C) Nuclear Division takes place.
(D) Reorganisation of all cell components takes place.

Open solution

Q46. NEET 2019 (Odisha)

Crossing over takes place between which chromatids and in which stage of the cell cycle?

(A) Non-sister chromatids of non-homologous chromosomes at Pachytene stage of prophase I
(B) Non-sister chromatids of non-homologous chromosomes at Zygotene stage of prophase I
(C) Non-sister chromatids of homologous chromosomes at Pachytene stage of prophase I
(D) Non-sister chromatids of homologous chromosomes at Zygotene stage of prophase I

Open solution

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Q47. NEET 2019 (Odisha)

After meiosis I, the resultant daughter cells have

(A) four times the amount of DNA in comparison to haploid gamete.

(B) same amount of DNA as in the parent cell in S phase.

(C) twice the amount of DNA in comparison to haploid gamete.

(D) same amount of DNA in comparison to haploid gamete.

Open solution

Q48. NEET 2019

Cells in G_0 phase :

(A) suspend the cell cycle

(B) terminate the cell cycle

(C) exit the cell cycle

(D) enter the cell cycle

Open solution

Q49. NEET 2019

The correct sequence of phases of cell cycle is :

(A) $S \rightarrow G_1 \rightarrow G_2 \rightarrow M$

(B) $G_1 \rightarrow S \rightarrow G_2 \rightarrow M$

(C) $M \rightarrow G_1 \rightarrow G_2 \rightarrow S$

(D) $G_1 \rightarrow G_2 \rightarrow S \rightarrow M$

Open solution

Q50. NEET 2018

The stage during which separation of the paired homologous chromosomes begins is

(A) zygotene.

(B) pachytene

(C) diplotene

(D) diakinesis

Open solution

Q51. NEET 2017

Which of the following options gives the correct sequence of events during mitosis?

(A) Condensation → Nuclear membrane disassembly → Arrangement at equator → Centromere division → Segregation → Telophase

(B) Condensation → Crossing over → Nuclear membrane disassembly → Segregation → Telophase

(C) Condensation → Arrangement at equator → Centromere division → Segregation → Telophase

(D) Condensation → Nuclear membrane disassembly → Crossing over → Segregation → Telophase

Open solution

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Q52. NEET 2017

Anaphase Promoting Complex (APC) is a protein degradation machinery necessary for proper mitosis of animal cell. If APC is defective in a human cell, which of the following is expected to occur?

Open solution

(A) Chromosomes will be fragmented.

(B) Chromosomes will not segregate.

(C) Recombination of chromosome arms will occur.

(D) Chromosomes will not condense.

Q53. NEET 2016 Phase 2

During cell growth, DNA synthesis takes place on

(A) S-phase **(B)** G_1-phase **(C)** G_2-phase **(D)** M phase

Open solution

Q54. NEET 2016 Phase 2

When cell has stalled DNA replication fork, which checkpoint should be predominantly activated?

(A) G_1/S **(B)** G_2/M **(C)** M **(D)** Both G_2/M and M

Open solution

Q55. NEET 2016 Phase 2

Match the stages of meiosis in column I to their characteristic features in column II and select the correct option using the codes given below.

Open solution

Column I		Column II	
A.	Pachytene	(i)	Pairing of homologous chromosomes
B.	Metaphase I	(ii)	Terminalisation of chiasmata
C.	Diakinesis	(iii)	Crossing-over takes place
D.	Zygotene	(iv)	Chromosomes align at equatorial plate

(A) (A) - (iii); (B) - (iv); (C) - (ii); (D) - (i)

(B) (A) - (i); (B) - (iv); (C) - (ii); (D) - (iii)

(C) (A) - (ii); (B) - (iv); (C) - (iii); (D) - (i)

(D) (A) - (iv); (B) - (iii); (C) - (ii); (D) - (i)

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Q56. NEET 2016 Phase 1

Spindle fibres attach on to

(A) centromere of the chromosome
(B) kinetosome of the chromosome
(C) telomere of the chromosome
(D) kinetochore of the chromosome.

Open solution

Q57. NEET 2016 Phase 1

Which of the following is not a characteristic feature during mitosis in somatic cells?

(A) Chromosome movement
(B) Synapsis
(C) Spindle fibres
(D) Disappearance of nucleolus

Open solution

Q58. NEET 2016 Phase 1

In meiosis crossing over is initiated at

(A) zygotene
(B) diplotene
(C) pachytene
(D) leptotene.

Open solution

Q59. AIPMT 2015

Arrange the following events of meiosis in correct sequence
(i) Crossing over
(ii) Synapsis
(iii) Terminalisation of chiasmata
(iv) Disappearance of nucleolus

(A) (i), (ii), (iii), (iv)
(B) (ii), (iii), (iv), (i)
(C) (ii), (i), (iv), (iii)
(D) (ii), (i), (iii), (iv)

Open solution

Q60. AIPMT 2015 Cancelled Paper

A somatic cell that has just completed the S phase of its cell cycle, as compared to gamete of the same species, has

(A) twice the number of chromosomes and four times the amount of DNA
(B) four times the number of chromosomes and twice the amount of DNA
(C) twice the number of chromosomes and twice the amount of DNA
(D) same number of chromosomes but twice the amount of DNA.

Open solution

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Q61. AIPMT 2015 Cancelled Paper

Select the correct option.

Column I			Column II	
A.	Synapsis aligns homologous chromosomes		(i)	Anaphase II
B.	Synthesis of RNA and protein		(ii)	Zygotene
C.	Action of enzymes recombinase		(iii)	G_2-phase
D.	Centromeres do not separate but chromatids move towards opposite poles		(iv)	Anaphase I
			(v)	Pachytene

Open solution

(A) (A) - (i); (B) - (ii); (C) - (v); (D) - (iv)

(B) (A) - (ii); (B) - (iii); (C) - (iv); (D) - (v)

(C) (A) - (ii); (B) - (i); (C) - (iii); (D) - (iv)

(D) (A) - (ii); (B) - (iii); (C) - (v); (D) - (iv)

Q62. AIPMT 2014

During which phase(s) of cell cycle, amount of DNA in a cell remains at 4C level if the initial amount is denoted as 2C?

(A) G_0 and G_1 **(B)** G_1 and S **(C)** Only G_2 **(D)** G_2 and M

Open solution

Q63. AIPMT 2014

In 'S' phase of the cell cycle

(A) amount of DNA doubles in each cell

(B) amount of DNA remains same in each cell

(C) chromosomes number is increased

(D) amount of DNA is reduced to half in each cell.

Open solution

Q64. AIPMT 2014

The enzyme recombinase is required at which stage of meiosis?

(A) Pachytene **(B)** Zygotene **(C)** Diplotene **(D)** Diakinesis

Open solution

Q65. NEET 2013 (Karnataka)

During meiosis I, the chromosomes start pairing at

(A) zygotene **(B)** pachytene **(C)** diplotene **(D)** leptotene.

Open solution

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Q66. NEET 2013 (Karnataka)

During the metaphase stage of mitosis, spindle fibres attach to chromosomes at

(A) kinetochore

(B) both centromere and kinetochore

(C) centromere, kinetochore and areas adjoining centromere

(D) centromere.

Open solution

Q67. NEET 2013 (Karnataka)

A stage of mitosis is shown in the diagram.
Which stage is it and what are its characteristics?



(A) Anaphase - Centromeres split and chromatids separate and start moving away.

(B) Late prophase - Chromosomes move to spindle equator.

(C) Metaphase - Spindle fibers attached to kinetochores, centromeres split and chromatids separate.

(D) Metaphase - Chromosomes moved to spindle equator, chromosomes made up of two sister chromatids.

Open solution

Q68. NEET 2013

The complex formed by a pair of synapsed homologous chromosomes is called

(A) bivalent (B) axoneme (C) equatorial plate (D) kinetochore.

Open solution

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Q69. NEET 2013

A stage in cell division is shown in the figure. Select the answer which gives correct identification of the stage with its characteristics.

Open solution



(A) Telophase Endoplasmic reticulum and nucleolus not reformed yet.

(B) Telophase Nuclear envelope reforms,Golgi complex reforms.

(C) Late anaphase Chromosomes move away from equatorial plate, Golgi complex not present.

(D) Cytokinesis Cell plate formed, mitochondria distributed between two daughter cells.

Q70. AIPMT 2012 Mains

Identify the meiotic stage in which the homologous chromosomes separate while the sister chromatids remain associated at their centromeres.

(A) Metaphase I (B) Metaphase II (C) Anaphase I (D) Anaphase II

Open solution

Q71. AIPMT 2012 Prelims

During gamete formation, the enzyme recombinase participates during

(A) metaphase I (B) anaphase II (C) prophase I (D) prophase II.

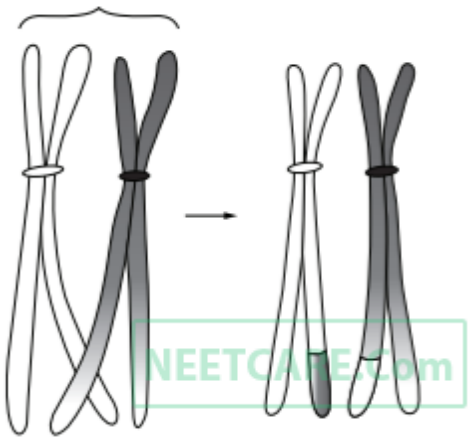
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Q72. AIPMT 2012 Prelims

Given below is the representation of a certain event at a particular stage of a type of cell division. Which is this stage?

Open solution



(A) Prophase of Mitosis
(B) Both prophase and metaphase of mitosis
(C) Prophase I during meiosis
(D) Prophase -II during meiosis

Q73. AIPMT 2011 Mains

At metaphase, chromosomes are attached to the spindle fibres by their

Open solution

(A) satellites
(B) secondary constrictions
(C) kinetochores
(D) centromeres.

Q74. AIPMT 2011 Prelims

Select the correct option with respect to mitosis.

Open solution

(A) Chromatids separate but remain in the centre of the cell in anaphase
(B) Chromatids start moving towards opposite poles in telophase
(C) Golgi complex and endoplasmic reticulum are still visible at the end of prophase.
(D) Chromosomes move to the spindle equator and get aligned along equatorial plate in metaphase.

Q75. AIPMT 2010 Prelims

During mitosis, ER and nucleolus begin to disappear at

Open solution

(A) late prophase
(B) early metaphase
(C) late metaphase
(D) early prophase.

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Q76. AIPMT 2010 Prelims

Which stages of cell division do the following figures A and B represent respectively?

Open solution

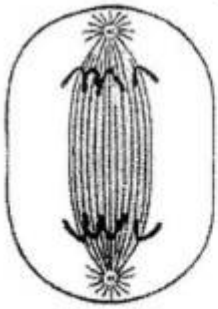


Fig. A



Fig. B

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(A) **A** → Prophase, **B** → Anaphase

(B) **A** → Metaphase, **B** → Telophase

(C) **A** → Telophase, **B** → Metaphase

(D) **A** → Late anaphase, **B** → Prophase

Q77. AIPMT 2009

Synapsis occurs between

Open solution

(A) *m*RNA and ribosomes

(B) spindle fibres and centromere

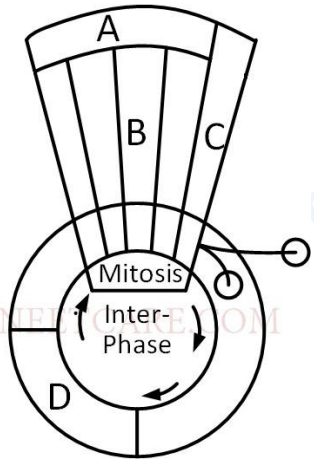
(C) two homologous chromosomes

(D) a male and a female gamete.

Q78. AIPMT 2009

Given below is a schematic break-up of the phases/stages of cell cycle. Which one of the following is the correct indication of the stage/phase in the cell cycle?

Open solution



(A) C - karyokinesis

(B) D - synthetic phase

(C) A - cytokinesis

(D) B - metaphase

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Q79. AIPMT 2005

At what stage of the cell cycle are histone proteins synthesized in a eukaryotic cell?

(A) During G_2 stage of prophase
(B) During S-phase
(C) During entire prophase
(D) During telophase

Open solution

Q80. AIPMT 2004

In the somatic cell cycle

(A) in G_1 phase DNA content is double the amount of DNA present in the original cell
(B) DNA replication takes place in S phase
(C) a short interphase is followed by a long mitotic phase
(D) G_2 phase follows mitotic phase.

Open solution

Q81. AIPMT 2004

If you are provided with root-tips of onion in your class and are asked to count the chromosomes, which of the following stages can you most conveniently look into?

(A) Metaphase (B) Telophase (C) Anaphase (D) Prophase

Open solution

Q82. AIPMT 2004

Which one of the following precedes reformation of the nuclear envelope during M phase of the cell cycle?

(A) Decondensation from chromosomes, and reassembly of the nuclear lamina
(B) Transcription from chromosomes, and reassembly of the nuclear lamina
(C) Formation of the contractile ring, and formation of the phragmoplast
(D) Formation of the contractile ring, and transcription from chromosomes.

Open solution

Q83. AIPMT 2004

Crossing over that results in genetic recombination in higher organisms occurs between

(A) sister chromatids of bivalent
(B) non-sister chromatids of a bivalent
(C) two daughter nuclei
(D) two different bivalents

Open solution

Q84. AIPMT 2002

Mitotic spindle is mainly composed of which protein?

(A) Actin (B) Myosin (C) Tubulin (D) Myoglobin

Open solution

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Q85. AIPMT 2002

Best material for the study of mitosis in laboratory is

(A) anther (B) root tip (C) leaf tip (D) ovary.

Open solution

Q86. AIPMT 2002

If a diploid cell is treated with colchicine then it becomes : -

(A) Tetraploid (B) Monoploid (C) Diploid (D) Triploid

Open solution

Q87. AIPMT 2002

What is the reason of formation of embryoid from pollen grain in tissue culture medium ?

(A) Double fertilization (B) Test tube culture
(C) Cellular totipotency (D) Organogenesis

Open solution

Q88. AIPMT 2000

Spindle fibre unite with which structure of chromosomes?

(A) Chromocentre (B) Chromomere (C) kinetochore (D) Centriole

Open solution

Q89. AIPMT 2000

In which stage of cell cycle, DNA replication occurs?

(A) G_1-phase (B) S-phase (C) G_2-phase (D) M-phase

Open solution

Q90. AIPMT 1998

Radioactive thymidine when added to the medium surrounding living mammalian cells gets incorporated into the newly synthesised DNA. Which of the following types of chromatin is expected to become radioactive if cells are exposed to radioactive thymidine as soon as they enter the S phase?

(A) Neither heterochromatin nor euchromatin but only the nucleolus
(B) Heterochromatin
(C) Euchromatin
(D) Both (b) and (c)

Open solution

Q91. AIPMT 1997

How many mitotic divisions are needed for a single cell to make 128 cells?

(A) 28 (B) 32 (C) 7 (D) 14

Open solution

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Q92. AIPMT 1997

During cell division in apical meristem, the nuclear membrane appears in

(A) telophase (B) cytokinesis (C) metaphase (D) anaphase

Open solution

Q93. AIPMT 1996

When paternal and maternal chromosomes change their materials with each other in cell division this event is called

(A) bivalent-forming (B) dyad-forming (C) synapsis (D) crossing-over

Open solution

Q94. AIPMT 1996

Which typical stage is known for DNA replication?

(A) S-phase (B) G_2-phase (C) metaphase (D) G_1-phase

Open solution

Q95. AIPMT 1996

Colchicine is an inhibitory chemical, which

(A) stops the functioning of centriole

(B) prevents attaching of centromeres with rays

(C) prevents the spindle formation in mitosis

(D) prevents the formation of equatorial plane

Open solution

Q96. AIPMT 1994

In a somatic cell cycle, DNA synthesis takes place in

(A) G_1 phase (B) prophase of mitosis

(C) S-phase (D) G_2 phase

Open solution

Q97. AIPMT 1994

Which of the following represents the best stage to view the shape, size and number of chromosomes?

(A) Prophase (B) Metaphase (C) Interphase (D) Telophase

Open solution

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Q98. AIPMT 1994

Which statement best explains the evolutionary advantage of meiosis?

(A) Meiosis is necessary for sexual reproduction

(B) Genetic recombinations are possible from generation to generation

(C) Meiosis alternates with mitosis from generation to generation

(D) The same genetic system is passed on from generation to generation

Open solution

Q99. AIPMT 1993

Meiosis II performs

(A) separation of sex chromosomes

(B) synthesis of DNA and centromere

(C) separation of homologous chromosomes

(D) separation of chromatids

Open solution

Q100. AIPMT 1992

Number of chromatids at metaphase is

(A) two each in mitosis and meiosis

(B) two in mitosis and one in meiosis

(C) two in mitosis and four in meiosis

(D) one in mitosis and two in meiosis

Open solution

Q101. AIPMT 1991

Mitotic anaphase differs from metaphase in possessing

(A) same number of chromosomes and same number of chromatids

(B) half number of chromosomes and half number of chromatids

(C) half number of chromosomes and same number of chromatids

(D) same number of chromosomes and half number of chromatids

Open solution

Q102. AIPMT 1991

In meiosis, the daughter cells differ from parent cell as well as amongst themselves due to

(A) segregation, independent assortment and crossing over

(B) segregation and crossing over

(C) independent assortment and crossing over

(D) segregation and independent assortment

Open solution

Q103. AIPMT 1990

Segregation of Mendelian factor (Aa) occurs during

(A) diplotene

(B) anaphase I

(C) zygotene/pachytene

(D) anaphase II

Open solution

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Q104. AIPMT 1989

A bivalent consists of

(A) two chromatids and one centromere
(B) two chromatids and two centromeres
(C) four chromatids and two centromeres
(D) four chromatids and four centromeres

Open solution

Q105. AIPMT 1988

Meiosis I is reductional division. Meiosis II is equational division due to

(A) pairing of homologous chromosomes
(B) crossing over
(C) separation of chromatids
(D) disjunction of homologous chromosomes

Open solution

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Answer Key

1. **D**	2. **D**	3. **D**	4. **C**	5. **D**
6. **D**	7. **D**	8. **A**	9. **D**	10. **B**
11. **A**	12. **D**	13. **C**	14. **B**	15. **D**
16. **D**	17. **D**	18. **C**	19. **A**	20. **B**
21. **A**	22. **B**	23. **A**	24. **C**	25. **D**
26. **D**	27. **B**	28. **D**	29. **C**	30. **C**
31. **B**	32. **B**	33. **D**	34. **D**	35. **B**
36. **D**	37. **B**	38. **C**	39. **A**	40. **C**
41. **A**	42. **A**	43. **A**	44. **A**	45. **A**
46. **C**	47. **C**	48. **C**	49. **B**	50. **C**
51. **A**	52. **B**	53. **A**	54. **B**	55. **A**
56. **D**	57. **B**	58. **C**	59. **D**	60. **A**
61. **D**	62. **C**	63. **A**	64. **A**	65. **A**
66. **A**	67. **D**	68. **A**	69. **B**	70. **C**
71. **C**	72. **C**	73. **C**	74. **D**	75. **A**
76. **D**	77. **C**	78. **B**	79. **B**	80. **B**
81. **A**	82. **C**	83. **B**	84. **C**	85. **B**
86. **A**	87. **C**	88. **C**	89. **B**	90. **C**
91. **C**	92. **A**	93. **D**	94. **A**	95. **C**
96. **C**	97. **B**	98. **B**	99. **D**	100. **A**
101. **D**	102. **A**	103. **B**	104. **C**	105. **C**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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